Thilan Dissanayaka

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Summary

Security Engineer with a software engineering background and professional experience in product security, vulnerability research, application security, and penetration testing. Experienced in assessing web applications and APIs, validating vulnerabilities, performing security-focused code review, analyzing authentication and authorization mechanisms, and working with engineering teams to remediate security issues. Strong foundation in Java, Go, C/C++, Linux, distributed systems, and WSO2 technologies, with an offensive-security mindset and interest in vulnerability research and secure software engineering.

Experience

Security Engineer **Jun 2025 – Present**
WSO2

- Perform security assessments and vulnerability research across enterprise products, with a focus on web applications, APIs, authentication, authorization, and access control.
- Conduct penetration testing and manual security testing to identify vulnerabilities that may not be detected by automated security scanners.
- Triage and validate vulnerability reports, assess exploitability and impact, and collaborate with engineering teams on remediation and security fixes.
- Perform security-focused source-code reviews and investigate vulnerability patterns including SSRF, XSS, access-control issues, injection vulnerabilities, authentication weaknesses, and insecure configurations.
- Analyze dependency vulnerabilities using software composition analysis and vulnerability intelligence tools, distinguishing exploitable product vulnerabilities from non-impacting or false-positive findings.
- Contribute to security advisory preparation, vulnerability severity assessment, remediation tracking, and coordinated disclosure processes.
- Perform security research across WSO2 technologies including API management, identity and access management, integration, and cloud platforms.
- Develop and evaluate security-testing approaches and tooling to improve the effectiveness of vulnerability discovery and validation.

Software Engineering Intern **Dec 2024 – June 2025**
WSO2

- Conducted research on token validation in edge-computing environments using AWS Lambda, EC2, Cloudflare, and Nginx.
- Developed a Ballerina CLI tool for translating smart contracts into Ballerina connectors using Java, the Ballerina Syntax Tree API, and Pico CLI.

Freelance Software developer **2012 – 2023**

- Developed commercial software including POS systems, Inventory management systems, HRM portals, Food ordering systems, Learning Management systems.
- Experienced in multiple tech stacks including PHP/MysQL, Java/SpringBoot, MERN stack, and Flutter.
- Built Web applications, Mobile apps, Desktop software and Web3 Integrations.

Education

Faculty of Engineering, University of Ruhuna **2020 – 2024**
BSc (Hons) Engineering in Electrical and Information Engineering

- Engineering degree covering Electrical, Electronic, Telecommunication, and Software Engineering disciplines, with opportunities for sub-specialization in later stages of the curriculum.
- Focused on software engineering, computer systems, and information security through coursework and engineering projects.

Technical Skills

- **Application Security:** Web Application Security, API Security, Penetration Testing, Vulnerability Research, Secure Code Review, Authentication, Authorization, Access Control, SSRF, XSS, SQL Injection, XXE.
- **Security Engineering:** Vulnerability Triage, Vulnerability Validation, CVSS, CWE, Dependency Security, Security Advisories, Static Analysis, Software Composition Analysis.
- **Security Tools:** Burp Suite, Nmap, sqlmap, Gobuster, Metasploit, GDB, YARA, Veracode, Mend, FOSSA, JFrog Xray.
- **Programming:** Java, Go, Python, C, C++, JavaScript, TypeScript, Rust, Ballerina, Bash.
- **Platforms & Technologies:** WSO2 API Manager, WSO2 Identity Server, Asgardeo, Choreo, Micro Integrator, Linux, Docker, Git, GitHub, AWS.

Projects and Research

CPU Simulation Software (cpusimulator.org)

- A research project focusing on simulating the entire computer CPU and memory areas such as Stack, Heap, etc.
- Capable of executing Intel assembly instructions on the CPU and examining the memory.
- Tools Used: React.js, Tailwind, GDB

AI-enabled Smart Grid Project

- Project in collaboration with LECO and the University of Ruhuna, funded by ADB.
- Actively contributed to the meter programming and server communication part.
- Tools Used: Embedded C, Nuvoton M031, Quectel BC92, DLMS protocol, TCP, MongoDB

Web3 tools for Ballerina programming language.

- A Web3 library and toolkit developed for Ballerina programming language.
- Technologies: Web3, Ballerina, Etherium, Java, and Solidity.

Rust based Supply chain attack detection tool

- A Supply chain attack prevention tool developed targeting NPM
- Technologies: Rust and NPM

Publications

Design and Implementation of a Web-Based CPU Simulator (In progress)
Thilan Dissanayaka

Blockchain based Real Estate marketplace
Thilan Dissanayaka, Manjitha Marasinghe, Sashiprabha Mihihansi

Highlights and Achievements

- Owns 11 CVEs including critical vulnerability findings.
- Top 5 performance award at WSO2.
- Won 2nd place from Sri Lanka at the IEEE R10 robotic competition.
- Chairman of the IEEE Computer Society, University of Ruhuna.

References

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